Rudder Trim Tab System – General Data

1. General

The rudder trim tab system is hand-operated from a trim tab handwheel, which is part of the trim control assembly mounted in the trim console to the right of the pilot's seat. Rotary movement of the handwheel is transmitted by a drum and cable system to a similar drum on a trim screw jack mounted in the upper portion of the rudder. From the screw jack the movement is transmitted by control rod to the rudder trim tab.

2. Description

The rudder trim tab system consists of the following components:

A. Trim Tab

The rudder trim tab is hinged to the trailing edge of the rudder upper portion and is connected to a screw jack by an adjustable control rod.

B. Trim Tab Screw Jack

The rudder trim tab screw jack converts rotary movement of the cable drum into linear motion to operate a push-pull rod. The screw jack is mounted on the front spar of the rudder structure in the upper portion of the rudder. The rudder trim tab screw jack is identical to the flap/elevator interconnect trim tab screw jack.

C. Rudder Trim Tab Handwheel

The rudder trim tab handwheel is hand-operated and is mounted on the trim controls console at the right of the pilot's seat. Rotation of the handwheel rotates a cable drum to drive the trim tab screw jack and an indicator pointer.

3. Operation

Rotation of the trim tab handwheel turns the cable drum, cables attached to the handwheel drum and the screw jack drum, move and rotate the screw jack. With rotation the screw jack either extends or retracts to operate a push-pull rod and so move the rudder trim tab. Extension of the screw jack moves the tab to the right (nose left trim), and retraction moves the tab to the left.

4. Adjustment/Test

A. Rig Rudder Trim Tab System (Mod 6/1219)

Refer to Figure 1 for details.

1) Rotate rudder trim tab handwheel until cables are arranged as in “Detail D”, and cable terminals are in position shown in “Detail C” of Figure 1.

2) Maintain position as in Para 4.A.1) and set rudder trim screw jack to 0.62 inch extension with cables arranged as shown in “Detail A” of Figure 1.

3) Maintaining positions as in Para 4.A.1) and Para 4.A.2), adjust and tension cables in rear fuselage. For cable tension values refer to 27-00-00, Flight Controls – General Data. Lock turnbuckles using locking clips.

Note The two cables from turnbuckles to screw jack are nylon covered, therefore tensiometer readings must be taken on the bare cables forward of the rear turnbuckles.

4) Disconnect connecting rod between screw jack and trim tab at trim tab attachment.

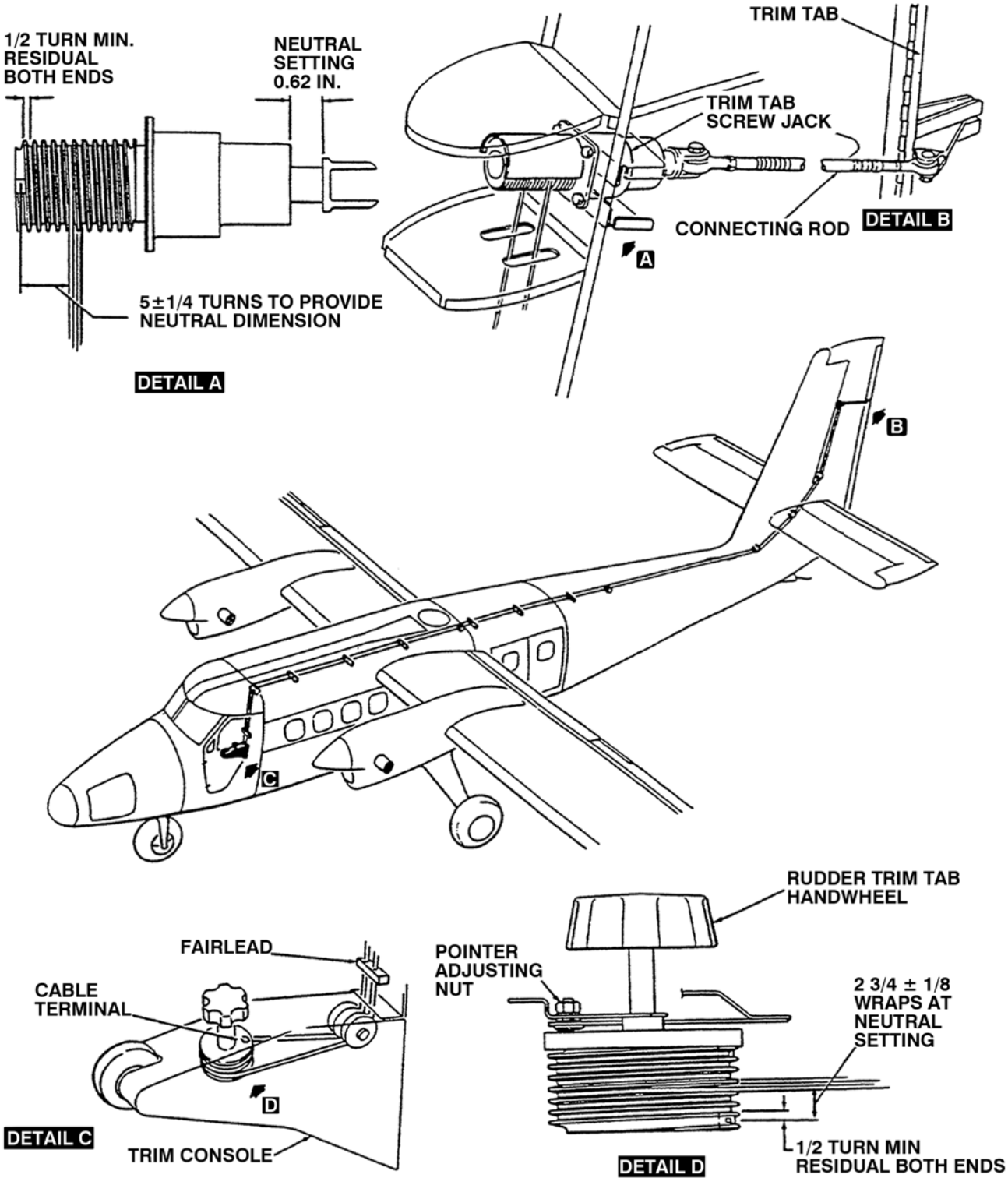
5) Set trim tab in neutral position by aligning trim tab and rudder trailing edges.

6) Maintaining position of screw jack as in Para 4.A.2), adjust and connect connecting rod to trim tab. Release nut securing handwheel pointer and align pointer with neutral position on trim console. Tighten pointer nut and connecting rod lock nut.

7) Rotate rudder trim handwheel to full nose left position, and to full nose right position. Check at extremities of travel that handwheel and screw jack drums have a minimum of half a turn of cable on each drum.

8) Check trim tab control surface clearances (refer to 27-00-00, Flight Controls – General Data).

9) Operate rudder and trim tab system and check for full and free range of movement.



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Figure 1

B. Alignment of Mechanical Rudder Trim Pointer and EFIS Rudder (Yaw) Trim Display

1) Install the control lock to set rudder to a neutral position.

2) Adjust Rudder Trim Tab pointer to full left position while monitoring the Electronic Flight Instrument System (EFIS) Rudder (Yaw) Trim Display.

3) Ensure that the EFIS display indicates full left at the same time the mechanical pointer indicates full left. Confirm Rudder Trim Tab is fully deflected right.

4) Adjust Rudder Trim Tab pointer to full right position while monitoring the EFIS display.

5) Ensure that the EFIS display indicates full right position at the same time the mechanical pointer indicates full right. Confirm Rudder Trim Tab is fully deflected left.

6) Adjust Rudder Trim Tab pointer to neutral position, while monitoring the EFIS display.

7) Ensure that the EFIS display indicates neutral position at the same time mechanical pointer indicates neutral. Confirm Rudder Trim Tab is at neutral.

8) If adjustment of the EFIS Rudder (Yaw) Trim Display Potentiometer is required (refer to 27-23-31, Rudder Trim Position Transmitter – General Data, Para 3.A.). If adjustment of the mechanical rudder trim display is required reference the mechanical trim system rigging, Para 4.A.

5. Inspection/Check

A. Free Play Check – Rudder Trim Tab

Refer to Figure 1 for details.

1) Operate rudder to neutral position.

2) Operate trim tab to neutral position.

3) Move trim tab manually and check that free play (backlash), measured at trailing edge, does not exceed 0.10 inch.

Note Measurement to be taken adjacent to control rod attachment point.